

Results of the 2nd International Workshop on the Impact of Agile Practices (ImpAct 2017)

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ABSTRACT

At present, agile development is a dominating development process in software engineering. Yet, due to different contexts, also agile methods require adaptations (e.g. Scrum-but). Since adaptation means adding, modifying or dropping some agile elements, it is important to know what the effects and importance of these elements are. Given the weak state of empirical evidence in this area, we initiated the workshop series on the Impact of Agile Practices (ImpAct). This paper provides a summary of the second workshop of this series, especially its lightning talks and discussions. The major outcomes include interesting observations such as negatively rated practices and contradicting experiences as well as follow-up activities ordered in a roadmap.

CCS CONCEPTS

• **Software and its engineering** → **Software development methods**;

KEYWORDS

Agile Development, Agile Practices, Impact, SPI.

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1 INTRODUCTION & MOTIVATION

Several common studies showed the importance and high usage of agile development approaches [1], such as Scrum. Nonetheless, the state of the art and practice in development processes showed that agile methods are often mixed with traditional processes [2],

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and it is often necessary to adapt agile methods to the particularities of specific organizational or project contexts [3]. Given the weak state of empirical evidence on the benefits and shortcomings of the plethora of agile practices at our disposal, we decided to concentrate the discussions on single agile practices (such as pair programming or using a product backlog) rather than on complete agile methods (such as Scrum).

Within the first Impact-workshop in 2015 [4] we covered a broad range of related topics, from foundational aspects captured in a meta model, over contextual factors important to consider when choosing specific issues, up to elaborate discussions about concrete impacts of single agile practices. In between that first workshop and the second one, we initiated a lightweight data collection phase to collect experts' perceptions on the impact of agile practices. To this end, we used interactive posters with sticky dots as introduced in [5]. Those posters were used in various venues where researchers and practitioners met and so we also employed them at this year's XP Conference 2017 where the second ImpAct workshop was co-located.



Figure 1 Data collection via interactive posters.

Figure 1 shows the poster as a means to collect impacts of agile practices on different quality attributes where conference attendees could add their own perception via stickers underlying a color coding (from strongly negative impacts to strongly positive ones). As for the XP Conference, we collected in total more than 500 data points which we used as an input for the workshop we held the last day of the one-week event. Please note that the impacts collected via this and all other posters can also be found on our website <http://impact.iese.fhg.de/data.php>.

We structured our workshop as follows. We began with two short invited lightning talks related to the overall topic of the impact of agile practices (Section 2). This was followed by a Lean Coffee session where we collected topics of interest from the participants. We prioritized them which then resulted in the topics used for the discussion (Section 3). Beside the discussion, we outlined ideas and plans for follow-up work (Section 4). The remainder of this paper is structured according to the workshop structure.

2 INVITED LIGHTNING TALKS

For starting and fostering the discussion on the impact of agile practices, we invited two speakers: Torgeir Dingsøy from SINTEF who talked about the practice Scrum-of-Scrums [6] for scaling agile approaches (Section 2.1), and Martin Kropp from FHNW Windisch who presented an excerpt of the current Swiss Agile Survey [7] related to the impacts of agile practices (Section 2.2). Both talks are detailed in the following.

2.1 Scrum-of-Scrums improves Communication

The first invited talk on Scrum-of-Scrums started with a statement of importance of coordination due to the fact that the main intension of this practice is to foster the communication between teams. Kraut and Streeter argue that while there is no single cause of the software crisis, a major contribution is the problem of coordinating activities while developing large software systems, which become much more difficult as project size and complexity increase [8]. After a short explanation of the practice in detail, the high usage and wide spread of the practice was shown by a common survey [1].

Furthermore, experiences were shared about whether Scrum-of-Scrum as a practice really works or not: Paasivaara et al. [9] present that “Scrum-of-Scrum meetings involving representatives from all teams were severely challenged: the audience was too wide to keep everybody interested [...] often ending up not reporting anything”. Additionally, [10] identified that “inter-group coordination becomes a major challenge when groups enjoy high levels of autonomy” which is or at least should be the case in agile development.

The presentation ended with a short discussion on different coordination modes [11], personal vs. impersonal and individual vs. group, all by giving different examples. One example for personal coordination is, of course, also the practice of Scrum-of-Scrums itself as a scheduled meeting.

2.2 Impact of Agile Practices in Switzerland

Martin Kropp offered insights into the ongoing data analysis of the last run of their survey conducted in Switzerland in 2016 on the impact of agile practices. Similar to [7] their results reported many companies using agile practices, but most of them used in fact not only agile practices but a combination with rather traditional approaches. This, again, strengthens our confidence in the existence of trends to use various practices for software

development. Besides this, the authors also provided insights into the application of single agile and plan-driven practices and their usage frequency.

Finally, and probably most important for us in the workshop, was a correlation analysis between the elicited practices and improvement goals, e.g. time to market, project visibility, or team productivity. This scope was similar to our poster results but emerged from a correlation analysis. The impacts are, thus, also to be seen bidirectional connections between practices and quality aspects.

3 FOLLOW-UP DISCUSSION

Directly following the two lightning talks, we began our discussion. We used the results of the poster-based data collection as a starting point. There, one major issue that raised our attention was that some cells either contained contradicting experiences (e.g. strong negative as well as strong positives) or they were mainly deemed negative, such as BurnDown Charts.

Next, we added further topics to be discussed in a Lean Coffee format including topics emerging from the interactive poster to methodological questions. After this, we prioritized these items based on dot voting (with three votes per participant). The resulting list including the votes is presented in the following:

- BurnDown Charts (3 votes)
- Scrum Meetings (1 vote)
- Product Quality (0 votes)
- Stress & Overtime (5 votes)
- Data Merging (2 votes)
- Reasoning (Why) (3 votes)
- Transfer to practice (0 votes)

As specified in the Lean Coffee concept, we started with the topic with the most votes and discussed that. Due to the fact that the first topic was interesting too all participants and we wanted to conclude the workshop with some future work aspects, we could only detail the aspect of stress and overtime. We summarize the corresponding discussions next.

3.1 Can Agile address Stress and Overtime and, if so, how?

First, we clarified that we define stress and overtime as workload (not in hours). Initially, we identified that every organizational change is stressful. That is, the introduction of agile development increases stress independently whether it is an evolutionary or revolutionary approach.

The first aspects that we discussed and identified covered specific agile practices that are supposed to support a better well-being:

- 40 hour week (XP)
- Mindfulness [12]
- Continuous Integration / Deployment / Delivery (in normal working hours)

Besides that, the second aspect deals with different context-issues that influence the stress and overtime level:

- Number of years working in agile

- Culture of the organization, project and/or team
- Work as well as tool environment

Together with context aspects, the issue of happiness appeared in the discussion due to the fact that stress and happiness appear to be opposites. This can be combined with the knowledge that happy developers are more productive [13][14] and probably less stressed.

Finally, we also discussed that stress can be positive. There is the concept of “eustress” which denotes stressors that are helping to succeed and create motivation and happiness. Only if the stressors become chronic, they can turn into negative effects. Agile practices such as sprints could support to have also regular stress relieve.

4 FUTURE WORK

Since our overall goal is to establish this workshop as a series, the discussion on future work is one of the major issues which we discussed and summarized at the end.

We came to the following issues which will be included in our future work, starting with very generic issues and finally coming to very detailed concrete ones:

- **Creating a Slack channel** for communication within the team or interested persons on that topics to proceed with all the other future work issues.
- **Summarizing the workshop** to spread our ideas and work in a community which is further fostered by this summary.
- **Writing a joint publication on challenges on collecting impacts** as a follow-up work to **Error! Reference source not found.** to discuss some of the aspects that we identified and discussed during the workshop, such as collection of certain context characteristics.
- **Establishing the workshop as a series** given the positive resonance. We plan to organize this workshop at subsequent conferences such as PROFES¹, XP², or ICSE³ due to their thematically fitting focus.
- **Establishing an online tool** (connected with our poster-data website) or **continuously running a survey** to ask participants two aspects: (1) First, to get more impacts on specific practices and (2) second on reasons for this given experienced impacts, such as context information or even wrong usage.
- **Developing hypotheses** at different levels of abstraction **from the existing poster data** (not only XP-data). For instance, are some practices used in a wrong manner? How does progress-tracking in practice work? Are agile practices (progress tracing) actionable? Out of such a huge list of hypothesis or questions that we can raise from the collected data we might be able to spread them as new PhD-totics.

As already mentioned, besides the more strategic aspects on the future work, we identified some concrete actions that might be related to the previous strategic issue:

- Identification of white spots in the existing data and checking whether this is the case (a) due to a lack of experiences by the participants in those specific areas or (b) because there are no impacts.
- Identification of differences or contradictions within the data to see which combinations of agile practice and characteristics need to be investigated, e.g., by giving contextual information.
- Question agile practices that are rated almost completely negatively to most or all improvement goals, such as Burn Down Charts during the XP conference.

Based on these different strategic and concrete items, which we derived during our intensive workshop discussions, we continue following the goal to fill the gap until we are going to conduct the next follow-up workshop on one of the mentioned conferences.

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¹ <http://profes-conferences.org/>

² <http://www.xp2018.org>

³ <http://www.icse2018.org>